

FutureMart Enterprise Data Architecture Assessment

Overview

This document evaluates the target-state enterprise data architecture of FutureMart following the merger of FashionMart and TrendyThreads.

The assessment identifies architectural strengths, improvement opportunities, scalability considerations, governance capabilities, and integration requirements across data ingestion, storage, processing, analytics, security, and business operations.

Enterprise

FutureMart Retail & E-Commerce Corporation

Architectural Assessment

Strengths

Serial No.	Strength	Description	Impact
1	Unified Omnichannel Data Architecture	FutureMart combines FashionMart’s physical retail data with TrendyThreads’ digital commerce data into one integrated architecture.	Provides a unified view of customers, sales, inventory, and operations across online and physical channels
2	Scalable Data Storage	The architecture uses centralized data warehouses, cloud storage, and real-time data stores to support structured and semi-structured data.	Supports business growth, larger transaction volumes, and future expansion.
3	Real-Time Data Integration	FutureMart uses API integration, data streaming, and ETL/ELT pipelines to connect online, in-store, inventory, and customer systems.	Improves data availability, reduces delays, and supports faster decision-making.
4	Advanced Analytics and Personalization	The architecture supports BI dashboards, predictive analytics, customer segmentation, and AI-based shopping recommendations.	Improves customer experience, sales performance, and marketing effectiveness.
5	Strong Data Governance and Security	FutureMart includes data governance, access control, encryption, compliance, audit logging, and data quality management.	Reduces compliance risks, protects customer data, and improves trust in enterprise data.

Areas for improvement

Serial No.	Area for Improvement	Description	Impact
1	Data Standardization	Data from FashionMart and TrendyThreads may use different formats, definitions, and structures.	Without standardization, reporting and analytics may be inconsistent across the merged enterprise.
2	Legacy System Modernization	Some FashionMart systems may still rely on traditional retail technologies and batch-oriented processing.	These systems can limit real-time visibility and slow down digital transformation.
3	Data Migration Complexity	Migrating data from separate systems into a unified FutureMart architecture requires careful mapping, cleansing, and validation.	Poor migration planning may cause data loss, duplication, or inconsistencies.
4	High Availability and Disaster Recovery	FutureMart needs strong failover, backup, and disaster recovery processes across both online and physical retail systems.	Improves business continuity and reduces downtime during failures or disruptions.
5	Governance Alignment Across Teams	The merged company must align data ownership, stewardship, access policies, and metadata standards across both organizations.	Prevents confusion, improves accountability, and supports long-term data quality.